

Important points to keep in mind

The three most important factors to be kept in mind are:

1. Make sure to get a clean input signal with no jammers. This can be done by placing the antenna facing the satellites and not near jammers such as the camera/processor/memory/ or other transmitter.
2. Make sure there is no thermal disturbance to the TCXO so that temperature compensation works well. This can be done by making sure not to mount the TCXO near a heat source and make sure there are thermal breaks in any power and ground planes in the PCB to prevent heat transfer from other components that use a lot of power.
3. Make sure to have a robust serial communication connection. It is amazing how such an old technology such as an UART can still be the cause of so many problems.

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process the gathered data to find out the current location. Modules with higher update frequencies perform better, but these are mostly used for aerospace, defence or other highly critical applications. Civilian applications usually require just one module with 1Hz update frequency.

The antenna size and type also affect the power requirements of your device. Typically, active antennae need more power than passive antennae. Within active antennae, there are high-performance antennae that provide a higher accuracy but are larger and more power consuming. Ultimately, it depends on the design engineer to balance the power requirements, device size and accuracy to create a successful product. One thing I learnt is that the power consumption figures mentioned in OEMV GPS cards do not include the power consumed by the antenna's low-noise amplifier (LNA).

GPS modules can use multiple channels at the same time to talk faster with the satellites and gather ephemeris data faster. However, with more channels comes the problem of increased power consumption. Typically, the design is such that more channels are used just for the initial fix, after which the additional channels are dropped.

While checking around for the lowest power-consuming GPS module, I came across LSI CXD5600GF and CXD5430 GNSS receivers from Sony that consume just 10 mW for the entire GNSS receiver LSI unit.

Decreasing time-to-first-fix. One solution to considerably reduce both

TTFF and power consumption of mobile communication devices is to utilise the wireless network and offload most of the work to the cloud. Nokia N95-1 is a pretty good example of this with A-GPS functionality added to the device over a software update. This software update alone decreased Time To First Fix (TTFF) for these phones by a big margin, making GPS functionality on the device a feasible option.

To understand this, you need to factor in the 30 seconds of full throttle work that the GPS module has to do usually, to simply acquire ephemeris data from the satellite. Once this data is received, the device has to go ahead and do some heavy number crunching to make sense of the data and calculate

the location precisely.

One alternative way to do this is to take useful information like satellite trajectories and earth elevation values from public databases, and post-process it on a remote server to calculate location.

"By using a coarse-time navigation technique and leveraging information that is already available on the Web, such as satellite ephemeris and Earth elevations, we show that 2 ms of raw GPS signals is enough to obtain a location fix," explains a paper on CO-GPS built through the wonderful Microsoft CLEO project.

You might be thinking why GPS is less preferable than your home network or Wi-Fi for receiving data. The reason is that GPS signals have a very low data rate. It is far more efficient to piggy-back on the already utilised data connection with your service provider rather than spending a couple of minutes sweet talking with the GPS satellite to try and get it to reveal where you are.

Accuracy. The accuracy of a GPS module primarily depends on the GPS module and the antenna. Active antennae are usually connected through a cable and have their own LNA, which gives them better performance. On

Role of various technologies

Global positioning system (GPS). GPS is a space-based satellite navigation system that provides location and time information to any receiver that can understand its signals. It requires unobstructed line-of-sight to at least four satellites, and is also undergoing a modernisation effort powered by satellites named GPSII.

World wide reference network (WWRN). WWRN, the AGPS server, delivers GPS orbit data that is valid for four hours. The data is sent through wireless communication networks from servers, and helps to eliminate the time to 'acquire satellites' directly. This technology has helped to reduce the TTFF from minutes to mere seconds.

Global navigation satellite system (GLONASS). The Russian equivalent of GPS, it currently has 21 satellites available in the skies. It greatly enhances the performance of GPS chips, which is why most modern devices have GLONASS support built-in.

Long-term orbit (LTO)4d. This technology delivers GPS data seven days in advance, and is used when the network might be unavailable for the next seven days. It builds on AGPS, with the difference that the ephemeris data is available for seven days instead of four hours. The main takeaway here is that by downloading LTO data, a user outside the home network can still get AGPS-like accuracy.

Galileo navigation satellite system (GNSS). It currently has just four satellites up, with 26 more to go up by 2019. However, engineers can execute full testing with the current four satellites. This system also offers better positioning at higher altitudes and has an integrated global search-and-rescue function that allows the user to send a distress signal to the satellite even if he is out of range of the home network.